SYNOPSIS

The Administration and Security System is a comprehensive database solution designed to streamline and manage critical organizational operations. This system focuses on two primary areas: administration and security, ensuring efficient management of employees, visitors, meal services, attendance, and leave records.

Purpose of the System

Organizations often face challenges in maintaining accurate records of employees, visitors, and daily operations while ensuring security protocols are adhered to. This database system addresses these challenges by offering a structured approach to managing administrative tasks, monitoring security, and enabling data-driven decision-making.

Key Features and Functionality

1. Employee Management:

- Maintain detailed records of employees, including their personal information, departmental assignments, and employment status.
- Keep track of active and inactive employees, simplifying workforce planning and analysis.

2. Visitor Management:

- Record visitor details, including their organization, contact information, purpose of visit, and check-in/check-out times.
- Link visitors to their host employees for accountability and tracking.

3. Meal Management:

- Schedule meals with options for breakfast, lunch, and dinner, including specific menus for different days.
- Track meal prices and allow seamless integration with employee and visitor meal records for billing or analysis.

4. Attendance Tracking:

- Monitor employee attendance with automatic calculation of work hours based on check-in and check-out times.

- Track attendance status (e.g., Present, Absent, On Leave) to simplify payroll and performance reviews.

5. **Leave Records:**
    - Maintain leave details for employees, including the type of leave, reason, and approval status.
    - Enable managers to approve or reject leave requests and generate reports on leave trends.

Security and Administration Benefits

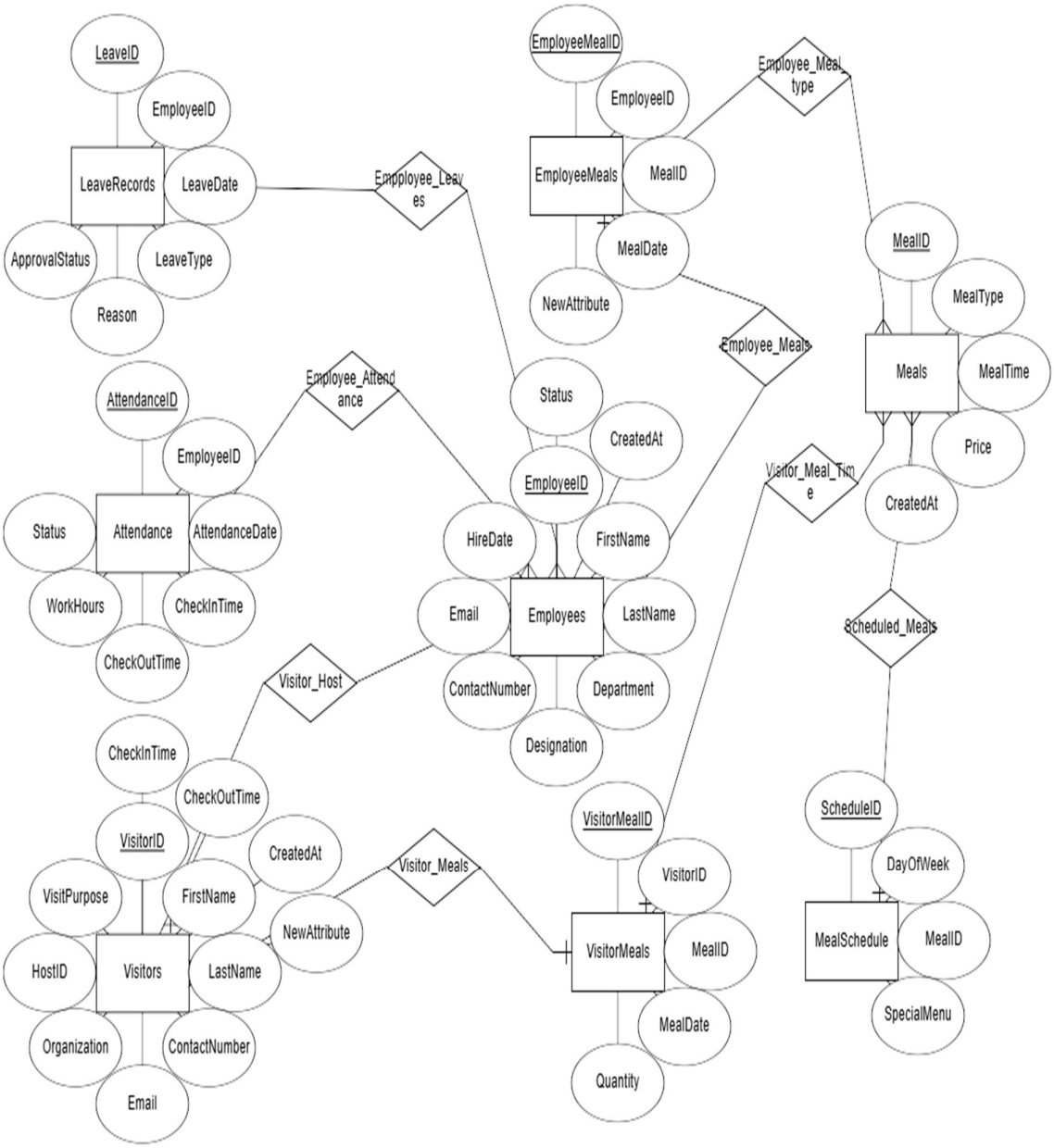
- Enhanced Data Security: By linking visitors to hosts and tracking their check-in/check-out times, the system ensures accountability and reduces unauthorized access.
- Operational Efficiency: Automating routine tasks like attendance tracking, leave management, and meal scheduling saves time and reduces human errors.
- Scalability: The system can be expanded to accommodate more employees, visitors, and features as the organization grows.
- Improved Reporting: The system provides detailed insights into attendance patterns, visitor history, and meal preferences, enabling data-driven decisions.

Why this Database is Essential

The Administration and Security System integrates multiple organizational functions into a single database, reducing redundancy and improving data accuracy. It provides a holistic view of operations, enabling managers to focus on core tasks without being bogged down by manual record-keeping. By combining administrative management with security protocols, this system ensures that both operational efficiency and workplace safety are maintained.

This project not only highlights the importance of database design in organizational management but also demonstrates how an efficient system can simplify complex workflows while maintaining a high level of security.

ER DIAGRAM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Email
Employees
LastName
ContactNumber
Department
Designation
Visitor_Host
CheckInTime
CheckOutTime
VisitorID
CreatedAt
VisitPurpose
FirstName
NewAttribute
HostID
Visitors
LastName
Organization
ContactNumber
Email
Visitor_Meals
VisitorMealID
VisitorID
VisitorMeals
MealID
MealDate
Quantity
ScheduleID
DayOfWeek
MealSchedule
MealID
SpecialMenu

LIST AND PICTURE OF ALL TABLES

Table: Employees

Field	Type	Description
EmployeeID	INT (Primary Key)	Unique employee ID.
FirstName	VARCHAR(50)	Employee's first name.
LastName	VARCHAR(50)	Employee's last name.
Department	VARCHAR(50)	Department name.
Designation	VARCHAR(50)	Job designation.
ContactNumber	VARCHAR(15)	Contact number.
Email	VARCHAR(100)	Email address.
HireDate	DATE	Hiring date.
Status	ENUM('Active', 'Inactive')	Employment status.
CreatedAt	TIMESTAMP	Record creation timestamp.

EmployeeID	FirstName	LastName	Department	Designation	ContactNumber	Email	HireDate	Status	CreatedAt
1	John	Doe	IT	Software Engineer	1234567890	john.doe@example.com	2022-01-15	Active	2024-11-27 10:36:18
2	Jane	Smith	HR	HR Manager	9876543210	jane.smith@example.com	2021-05-20	Active	2024-11-27 10:36:18
3	Mike	Johnson	Finance	Accountant	4567891230	mike.johnson@example.com	2023-03-01	Active	2024-11-27 10:36:18
4	Emily	Brown	Marketing	Marketing Specialist	7891234560	emily.brown@example.com	2020-07-10	Inactive	2024-11-27 10:36:18
5	John	Doe	HR	Manager	1234567890	john.doe@example.com	2023-01-15	Inactive	2024-11-29 13:15:10
6	Jane	Smith	IT	Developer	9876543210	jane.smith@example.com	2022-11-20	Active	2024-11-29 13:15:10
7	Robert	Brown	Finance	Analyst	4567891230	robert.brown@example.com	2021-08-10	Inactive	2024-11-29 13:15:10
8	Alice	Johnson	Admin	Assistant	7891234560	alice.johnson@example.com	2024-02-01	Active	2024-11-29 13:15:10
9	Michael	Clark	HR	Recruiter	3216549870	michael.clark@example.com	2023-07-25	Active	2024-11-29 13:15:10
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

VisitorID	FirstName	LastName	ContactNumber	Email	Organization	HostID	VisitPurpose	CheckInTime	CheckOutTime	CreatedAt
1	Robert	Taylor	1112223333	robert.taylor@abc.com	ABC Corp	1	Business Meeting	2024-11-25 09:00:00	2024-11-25 11:00:00	2024-11-27 10:36:18
2	Laura	Martin	2223334444	laura.martin@xyz.com	XYZ Ltd	2	Job Interview	2024-11-26 10:00:00	NULL	2024-11-27 10:36:18
3	Emily	Davis	1237894560	emily.davis@company.com	ABC Corp	1	Business Meeting	2024-11-28 09:00:00	2024-11-29 14:00:00	2024-11-29 13:15:19
4	David	Wilson	4561237890	david.wilson@client.com	XYZ Ltd	2	Project Discussion	2024-11-28 14:00:00	NULL	2024-11-29 13:15:19
5	Sophia	Taylor	7894561230	sophia.taylor@partner.com	Partner Co	3	Sales Pitch	2024-11-29 11:00:00	2024-11-29 12:00:00	2024-11-29 13:15:19
6	James	Anderson	3219876540	james.anderson@guest.com	Freelancer	4	Consultation	2024-11-29 15:00:00	NULL	2024-11-29 13:15:19
7	Olivia	Martinez	6543219870	olivia.martinez@org.com	Non-Profit Org	1	Workshop	2024-11-27 10:00:00	2024-11-27 12:30:00	2024-11-29 13:15:19
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

MealID	MealType	MealTime	Price	CreatedAt
1	Breakfast	08:00:00	50.00	2024-11-27 10:36:18
2	Lunch	13:00:00	15.00	2024-11-27 10:36:18
3	Dinner	20:00:00	150.00	2024-11-27 10:36:18
5	Lunch	13:00:00	15.00	2024-11-29 13:15:19
6	Dinner	19:30:00	12.50	2024-11-29 13:15:19
7	Lunch	12:45:00	15.00	2024-11-29 13:15:19
8	Breakfast	09:00:00	6.00	2024-11-29 13:15:19
NULL	NULL	NULL	NULL	NULL

EmployeeMealID	EmployeeID	MealID	MealDate
1	1	2	2024-11-28
2	2	1	2024-11-28
3	3	3	2024-11-29
NULL	NULL	NULL	NULL

ScheduleID	DayOfWeek	MealID	SpecialMenu
1	Monday	1	Pancakes, Coffee
3	Monday	3	Pasta, Garlic Bread
4	Tuesday	1	Omelette, Tea
5	Tuesday	2	Beef Steak, Mashed Potatoes, Veggies
6	Tuesday	3	Pizza, Soft Drink
7	Monday	1	Vegan Options
8	Tuesday	2	Grilled Specials
9	Wednesday	3	Vegetarian Special
11	Friday	5	Burger Feast
12	Saturday	2	Weekend Brunch
13	Sunday	3	Chef's Choice
NULL	NULL	NULL	NULL

AttendanceID	EmployeeID	AttendanceDate	CheckInTime	CheckOutTime	WorkHours	Status
1	1	2024-11-25	2024-11-25 09:00:00	2024-11-25 17:00:00	8.00	Present
2	2	2024-11-25	2024-11-25 09:15:00	2024-11-25 17:15:00	8.00	Present
3	3	2024-11-25	NULL	2024-11-29 17:30:00	NULL	On Leave
4	1	2024-11-28	2024-11-28 08:00:00	2024-11-28 16:00:00	8.00	Present
6	3	2024-11-28	NULL	NULL	NULL	Absent
7	4	2024-11-27	2024-11-27 08:30:00	2024-11-27 15:30:00	7.00	Present
8	5	2024-11-29	2024-11-29 10:00:00	2024-11-29 18:00:00	8.00	Present
NULL	NULL	NULL	NULL	NULL	NULL	NULL

LeaveID	EmployeeID	LeaveDate	LeaveType	Reason	ApprovalStatus
1	3	2024-11-25	Sick Leave	Fever and cold	Approved
2	4	2024-11-26	Casual Leave	Family function	Pending
3	1	2024-11-25	Sick Leave	Flu Symptoms	Approved
4	2	2024-11-26	Casual Leave	Family Emergency	Pending
6	4	2024-11-28	Unpaid Leave	Personal Work	Rejected
7	5	2024-11-29	Sick Leave	Doctor Appointment	Pending
NULL	NULL	NULL	NULL	NULL	NULL

1.Employees Table

SELECT * FROM Employees WHERE Status = 'Active';

SELECT * FROM Employees WHERE Department = 'HR';

SELECT * FROM Employees WHERE HireDate > DATE_SUB(CURDATE(), INTERVAL 1 YEAR);

UPDATE Employees SET Status = 'Inactive' WHERE EmployeeID = 5;

SELECT Department, COUNT(*) AS EmployeeCount FROM Employees GROUP BY Department;

SELECT * FROM Employees ORDER BY HireDate DESC LIMIT 1;

SELECT CONCAT(FirstName, ' ', LastName) AS FullName FROM Employees;

DELETE FROM Employees WHERE EmployeeID = 10;

SELECT * FROM Employees WHERE FirstName LIKE '%John%';

SELECT * FROM Employees;

2.Visitors Table

SET SQL_SAFE_UPDATES = 0;

SELECT * FROM Visitors WHERE DATE(CheckInTime) = CURDATE();

SELECT * FROM Visitors WHERE HostID = 1;

SELECT CONCAT(FirstName, ' ', LastName) AS FullName, Organization FROM Visitors;

SELECT Organization, COUNT(*) AS VisitorCount FROM Visitors GROUP BY Organization;

UPDATE Visitors SET CheckOutTime = '2024-11-29 14:00:00' WHERE VisitorID = 3;

DELETE FROM Visitors WHERE DATE(CheckOutTime) = DATE_SUB(CURDATE(), INTERVAL 1 DAY) AND VisitorID IS NOT NULL;

SELECT *, TIMESTAMPDIFF(HOUR, CheckInTime, CheckOutTime) AS Duration FROM Visitors WHERE TIMESTAMPDIFF(HOUR, CheckInTime, CheckOutTime) > 2;

SELECT * FROM Visitors WHERE CheckOutTime IS NULL;

SELECT * FROM Visitors WHERE VisitPurpose LIKE '%Meeting%';

SELECT * FROM Visitors;

3. Meals Table

SELECT * FROM Meals WHERE Price < 10.00;

SELECT MealType, COUNT(*) AS MealCount FROM Meals GROUP BY MealType;

SELECT * FROM Meals ORDER BY Price DESC LIMIT 1;

SELECT * FROM Meals WHERE MealTime > '18:00:00';

UPDATE Meals SET Price = 15.00 WHERE MealType = 'Lunch';

DELETE FROM Meals WHERE MealID = 4;

SELECT * FROM Meals WHERE MealTime < '12:00:00';

SELECT AVG(Price) AS AveragePrice FROM Meals;

SELECT DISTINCT MealType FROM Meals;

SELECT * FROM Meals;

4.Employee Meals Table

SELECT * FROM EmployeeMeals WHERE EmployeeID = 1;

SELECT EmployeeID, COUNT(*) AS MealCount FROM EmployeeMeals GROUP BY EmployeeID;

SELECT * FROM EmployeeMeals WHERE MealDate = '2024-11-28';

SELECT em.EmployeeID, m.MealType FROM EmployeeMeals em JOIN Meals m ON em.MealID = m.MealID WHERE em.EmployeeID = 1;

SELECT em.EmployeeID, SUM(m.Price) AS TotalCost FROM EmployeeMeals em JOIN Meals m ON em.MealID = m.MealID WHERE em.EmployeeID = 1;

DELETE FROM EmployeeMeals WHERE EmployeeMealID = 5;

SELECT EmployeeID, MealDate, COUNT(*) AS MealCount FROM EmployeeMeals GROUP BY EmployeeID, MealDate HAVING MealCount > 2;

UPDATE EmployeeMeals SET MealDate = '2024-11-29' WHERE EmployeeMealID = 3;

SELECT m.MealType, COUNT(*) AS Frequency FROM EmployeeMeals em JOIN Meals m ON em.MealID = m.MealID GROUP BY m.MealType ORDER BY Frequency DESC LIMIT 1;

SELECT * FROM EmployeeMeals;

5.Meal Schedule Table

SELECT * FROM MealSchedule WHERE DayOfWeek = 'Monday';

SELECT ms.DayOfWeek, m.MealType, m.MealTime FROM MealSchedule ms JOIN Meals m ON ms.MealID = m.MealID;

SELECT DayOfWeek, COUNT(*) AS MealCount FROM MealSchedule GROUP BY DayOfWeek;

UPDATE MealSchedule SET SpecialMenu = 'Vegetarian Special' WHERE DayOfWeek = 'Wednesday';

DELETE FROM MealSchedule WHERE ScheduleID = 2;

SELECT * FROM MealSchedule WHERE DayOfWeek IN ('Saturday', 'Sunday');

SELECT DayOfWeek FROM MealSchedule WHERE MealID = 1;

SELECT * FROM MealSchedule ORDER BY FIELD(DayOfWeek, 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday');

SELECT DISTINCT SpecialMenu FROM MealSchedule;

SELECT * FROM MealSchedule;

6.Attendance Table

SELECT * FROM Attendance WHERE EmployeeID = 1;

SELECT * FROM Attendance WHERE AttendanceDate = CURDATE();

SELECT * FROM Attendance WHERE AttendanceDate = '2024-11-28' AND Status = 'Present';

UPDATE Attendance SET CheckOutTime = '2024-11-29 17:30:00' WHERE AttendanceID = 3;

SELECT * FROM Attendance WHERE WorkHours > 8;

SELECT Status, COUNT(*) AS StatusCount FROM Attendance WHERE AttendanceDate = '2024-11-28' GROUP BY Status;

SELECT SUM(WorkHours) AS TotalWorkHours FROM Attendance WHERE EmployeeID = 1 AND MONTH(AttendanceDate) = 11 AND YEAR(AttendanceDate) = 2024;

DELETE FROM Attendance WHERE AttendanceID = 5;

SELECT * FROM Attendance WHERE CheckOutTime IS NULL AND AttendanceDate = CURDATE();

SELECT * FROM Attendance;

7.LeaveRecords Table

SELECT * FROM LeaveRecords WHERE EmployeeID = 1;

SELECT * FROM LeaveRecords WHERE ApprovalStatus = 'Approved';

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SELECT LeaveType, COUNT(*) AS LeaveCount FROM LeaveRecords GROUP BY LeaveType;

SELECT * FROM LeaveRecords WHERE MONTH(LeaveDate) = 11 AND YEAR(LeaveDate) = 2024;

UPDATE LeaveRecords SET ApprovalStatus = 'Approved' WHERE LeaveID = 3;

SELECT * FROM LeaveRecords WHERE Reason LIKE '%medical%';

SELECT COUNT(*) AS PendingLeaves FROM LeaveRecords WHERE EmployeeID = 1 AND ApprovalStatus = 'Pending';

DELETE FROM LeaveRecords WHERE LeaveID = 5;

SELECT EmployeeID FROM LeaveRecords WHERE LeaveDate = '2024-11-28';

SELECT * FROM LeaveRecords;
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